

B. SUPPLEMENTARY GEOLOGY TABLES AND THE POSEIDON SURFACE ALBEDO DATABASE

Table A1. Supplementary Geology Glossary

Word (1)	Definition (2)	Exoplanet Context (3)
Mineral	Inorganic element or compound with a orderly internal structure and characteristic chemical composition. See succinct introduction on minerals, Vaughan (2014) .	An exoplanet’s observed disc-integrated dayside being composed of pure minerals is unlikely, though the surface can have rocks with large mineral inclusions (e.g., phenocrysts). Depending on the rock, distinct spectral signatures of specific minerals (like quartz in granite) can be readily detected.
Rock	An aggregate of one or more minerals. Rocks are determined by their mineralogy and bulk chemical composition, where resultant spectra is determined by mineralogy (i.e., two rocks of the same chemical composition can have different spectra due to different mineralogy (Paragas et al. 2025)). See succinct introduction on rocks, Zalasiewicz (2016) (Z16).	If the observed dayside of a rocky planet is completely cooled and solid, we are likely to observe spectral signatures of specific rocks (or, a conglomeration of rocks). The resulting color and spectral features of a rock is largely given by its mineral composition, though typically minerals do not spectroscopically mix in a linear fashion (Hu et al. 2012 ; Ehlmann & Mustard 2012).
Igneous Rocks	Rocks formed from the cooling of lava (extrusive/volcanic) or magma (intrusive/plutonic). They are the most common rock found on the surface of rocky planets in the Solar System, and are the first to form as a rocky planet cools from its initial formation state (magma ocean) and resurface a planet’s surface from subsequent partial melts of mantle material. ^{Z16}	With the exception of Earth, igenous rocks are the most common surface rock for Solar System rocky planets. The surfaces of currently observable rocky exoplanet daysides will be hot enough that we expect them to be molten or cooled molten, they therefore compose a majority of our database and the variety of processes that form them are detailed below in this table.
Sedimentary Rocks	Rocks formed from accumulation of pre-existing rocks over time. They are the most common rocks on Earth’s surface, due to its unique processes like plate tectonics (that pushes portions of the crust high enough into the atmosphere to erode), atmospheric processes (winds), surface water (oceans and rivers), and glacial processes. Some common sediments that form from igenous rocks include clays, muds, sand (mostly composed of quartz), and gravel; salts and calcium carbonates are also considered sedimentary and precipitate directly out of ocean water. Common rocks that form from the lithication of sediments are mudrocks (e.g., shale), sandstone, and limestone. Surface water and limestones have an inverse relationship with atmospheric carbon dioxide in the Solar System: Earth’s carbon dioxide is stored in limestone while Venus, which has no limestone, has a carbon dioxide dominated atmosphere (Hunten 1993). Sedimentary rocks can indicate the presence of past (sedimentary rocks have been found in proposed river basins on Mars (Lewis & Aharonson 2014)) or present liquid water, or just any process (like atmospheric erosion) that can create and accumulate sediment over time (i.e., dunes formed by winds on Mars, or see potential Venus sedimentary processes (Carter et al. 2023)). ^{Z16}	Sedimentary rocks require atmospheres and/or liquid water to form. Sedimentary rocks can form in hot and dense atmospheric environments (similar to Venus) that could be present in currently observable exoplanets. They could also possibly be present on airless rocky exoplanets that once housed a substantial atmospheres and/or surface water. The database presented in this work includes a few clays and salts.

Table A1 *continued*

Table A1 (continued)

Word (1)	Definition (2)	Exoplanet Context (3)
Metamorphic Rocks	Rocks formed from transformation of other rocks due to high heat and/or pressure, or hydrothermally. In Earth's traditional rock cycle, sedimentary rocks formed from igneous rocks get buried over time (i.e., in mountain belts) where, due to higher pressures and temperatures (regional metamorphism) mica minerals form from mudstones (that then can metamorphize a variety of rocks, like slate, schist, and gneiss, where gneiss is nearly identical in composition to the igneous rock granite), sandstone forms quartzite, and limestone forms marble, before finally melting into a magma to form new igneous rocks. High-pressure, low-temperature metamorphism that occurs at subduction zones in the ocean (hydrothermal) can form the mineral serpentine. Contact metamorphism occurs when rock is heated and crystallized due to heating from nearby magma. Impact metamorphism is where large impacts form glasses and high-pressure minerals. Metamorphic rocks are present on the surface of Earth due to geologic uplift (via plate tectonics) and subsequent erosion of overlying layers. ^{Z16} Metamorphic rocks could be the natural form of rocks of a surface with an overlying thick and high temperature atmosphere, as in the case of Venus (though, chemical alteration via sulfuric acid might dominate Venus's metamorphic processes) (Barsukov et al. 1982; Semprich et al. 2025). Hydrothermal and impact metamorphic rocks have been found on Mars (Hazen et al. 2023).	For rocky, airless exoplanets with magma oceans at their substellar point, the solidified rock around the magma ocean might be primarily in the form of metamorphic rocks from contact metamorphism with the neighboring magma ocean. On close-in rocky exoplanets, a form of high-temperature, low-pressure metamorphism might occur on the dayside due to incident irradiation (Chao et al. 2021). Exoplanets with Venusian like atmospheres might have chemical alteration of the surface. Evidence of past bombardment can form specific rocks via impact metamorphism. Past surface water can also be probed via the presence of hydrothermal metamorphic minerals (like serpentine). While these are an incredibly important class of rock, the database presented in this work does not include any explicit metamorphic rocks, but does include some rocks with serpentine.
Intrusive vs Extrusive	Many igneous rocks have different names for intrusive vs extrusive variants of rocks with identical chemical compositions (and therefore, similar absorption features in their spectra). The main difference between the two variants are their textures, where intrusive rocks are typically coarse grained due to slow-cooling (which usually occurs underground, which allows large crystals to grow) while extrusive rocks are typically fine-grained due to rapid cooling (usually due to lava interfacing with an atmosphere or surface water, causing it to rapidly cool). ^{Z16}	The difference in texture can result in differences in observed albedos and spectra (Paragas et al. 2025; Hammond et al. 2025). Where relevant, we will indicate when two geological categories in our database are an intrusive/extrusive pair (examples being: granite/rhyolite, diorite/andesite, gabbro/basalt), though we note that care must be taken when applying Earth-like definitions of intrusive vs extrusive to exoplanet applications.
TAS Diagram	The Total Alkali-Silica (TAS) diagram categorizes extrusive igneous rocks based on alkali metal oxide ($\text{Na}_2\text{O}+\text{K}_2\text{O}$) and SiO_2 content to distinguish whether they formed from alkaline-rich/poor or silica-rich/poor magma. For a textbook review of magma petrology, see Winter (2014) (W14) or Philpotts & Ague (2022).	While not directly applicable to exoplanets, many rocks are explicitly classified on this diagram. When relevant, we will indicate when a geological category in our database is a specific rock category on the TAS Diagram.
Alkaline Content in Magma	On Earth, alkali-rich magma is thought to form deep in the mantle, and found at areas of continental rifts and hotspots, while alkali-poor magma is thought to form from shallow melting of the mantle and is found at mid-ocean ridges where it cools to form the most common igneous rock on Earth (subcategorized as tholeiitic (iron enriched) and calc-alkaline (no iron enrichment)). ^{W14}	Alkali oxides, by themselves, do not have strong mid-infrared features, and therefore are not readily probed with JWST. However, feldspars (which contain alkalis + silica) which compose many igneous rocks can be probed by their mid-infrared features.
Silica Content in Magma	The silica content of magma is mostly determined by the age of magma and its degree of cooling (as magma cools, dense crystallized refractory material sinks and the remaining magma is enriched with silica). It is also used to determine whether or not a magma is mantle-derived (basaltic) or a crustal (granitic) melt. The silica content of magma strongly determines the viscosity (higher silica content results in a higher viscosity/more thick magma), and therefore determines how volcanoes erupt. ^{W14} Silica content can be used to further categorize rocks into ultramafic, mafic, intermediate, and felsic categories (see Table A3).	Silica content is the most accessible characteristic of rocky exoplanet surfaces that can be derived from spectroscopy with JWST due to strong mid-infrared Si-O absorption band and the Christiansen Feature. In our database, we choose to focus on silica content for this reason. Since silica content gives a wealth of degenerate information about the surface, such as initial formation abundances and degree of partial melting, care must be taken in interpreting it.

Table A1 continued

Table A1 (continued)

Word (1)	Definition (2)	Exoplanet Context (3)
Mantle Material	The mantle for evolved planets (with a fully cooled crust, and a fractionated core) is the middle region of a terrestrial body (in contrast, for lava worlds without a cooled crustal region the observable surface is often referred to as the mantle). Surface geology is intrinsically linked to a planet's mantle chemistry, wherein partial melts of mantle material cools and is reprocessed over time to form the surface of a planet. Typical mantle mineralogy and composition can be modeled from stellar compositions (see §5.1). On Earth, 90% of the upper mantle is composed of olivine, orthopyroxene, and clinopyroxene minerals that form mostly peridotite (and some pyroxenite) rocks. ^{Z16}	Earth-like upper mantle rocks (composed of olivines and pyroxenes) are predicted to be the dominant mantle material for exoplanets from models using the <i>Hypatia Catalog</i> (Putirka & Rarick 2019; Wang et al. 2022). However, more exotic mantle chemistry has been observed from white-dwarf pollution measurements where quartz and periclase saturation can instead occur (Putirka & Xu 2021). While the detection of surface composition can hint at a parent mantle composition, white dwarf pollution measurements are the only method by which to directly measure the mantle compositions of exoplanets (Xu & Bonsor 2021).
Partial Melts	As a terrestrial body cools from its initially molten state, refractory material with high melting points (i.e., olivines, pyroxenes) begin to solidify and remove themselves from the melt mixture, changing the bulk composition of the remaining melt. If the solidified material is removed from the system (i.e. sequestered deeper in the body via differentiation), the resulting melt will cool into a rock with a different composition from the original melt. The reverse process is also true, where as solidified mantle rocks are heated, volatile material begins to melt. Both processes above play a large role in determining the surface composition ^{Z16}	Partial melts of solid mantle material in Earth (i.e., ultramafic peridotites) produce more silica-rich magmas that cool to form mafic basalts. This pattern extends to all Solar System rocky planets, and is predicted to extend to most exoplanets. In general, more silica-rich material is more volatile and is first to melt (last to solidify), which is important to take into account when predicting the surface composition of planets with more exotic mantle compositions.
Melt fraction	The percentage of a rock that has melted from an original rock composition, where the resulting partial melt is typically more Si-rich than the original rock composition.	For lava worlds, the composition of the observable lava on the dayside will have a composition that is a melt fraction of the mantle. The quenched glasses from Fortin et al. (2022) represent the resulting melt chemistry of a mantle that has been melted a certain percentage and then rapidly quenched (to retain the internal structure of the magma). Magma composition, like solidified rocks, can be probed spectroscopically.
Oxygen Fugacity	fO_2 quantifies the amount of available oxygen to take part in oxidizing (loss of electrons/gaining oxygen) or reducing (gain of electrons/losing oxygen) chemical reactions in the interior of a planet. For internal geology, fO_2 is proportional to Fe^{+3} (ferric) / Fe^{+2} (ferrous) abundances. Low fO_2 favors metallic iron (Fe) and wüstite (FeO), while high fugacity favors hematite (Fe ₂ O ₃) (Cottrell et al. 2025). Oxygen fugacity controls the amount of interior metallic iron (and therefore the size of the iron core) (Unterborn & Panero 2017), the ease by which rock melts to produce magma (with higher fugacities resulting in a lower melting point) (Lin et al. 2021), and which molecular species are out-gassed (Gaillard et al. 2021) (i.e., H ₂ vs. H ₂ O, CO vs CO ₂ , H ₂ S vs SO ₂). In the Solar System, Earth's mantle is the most oxidized while Mars and the Moon are more reduced (where, Deng et al. (2020) argue that deeper magma oceans on larger planets naturally result in a more oxidized surface). While the above concerns mantle oxygen fugacity, the atmosphere of a planet can also reduce or oxidize crustal material (see Oxidized vs Reduced Surface in Table A3).	While the oxygen fugacity of an exoplanet's mantle cannot be directly known, many studies have explored how detecting specific atmospheric compositions can be tied to a mantle's oxygen fugacity (e.g., Drant et al. 2025; Perez et al. 2025). White dwarf pollution studies have shown that many exoplanets probably form with Earth-like oxygen fugacities (Doyle et al. 2019). While the amount of reactive oxygen in the mantle is directly tied to its formation conditions, the actual mantle oxygen fugacity can change depending on a planet's mineralogy (which depends on the initial Mg/Si from formation), wherein suppressed formation of a mineral like pyroxene can inherently increase mantle oxygen fugacity (Guimond et al. 2023) and change which molecular species are outgassed.

Table A1 continued

Table A1 (*continued*)

Word (1)	Definition (2)	Exoplanet Context (3)
Crustal Material	For terrestrial planets with a solid and cooled surface, the crust represents the top-most layer of the planet. The resulting bulk composition of the crust changes is a strong function of time from a planet's initial formation and is a complex result of cooling processes (like differentiation), resurfacing processes (like plate tectonics, volcanism, and large impacts), and alteration processes (like aqueous alteration and atmospheric oxidation). Here, we include the the crustal definitions of Condie (2022) (C22) that outline the types of crusts in the Solar System.	For exoplanets with cooled crusts, their dayside surface composition will be a strong probe of past and current geologic processes.
Primitive Crust	The crust that forms first from planetary accretion, which is common on smaller asteroids and comets, but rarely survives on larger bodies (due to their large heating during formation, which forms a magma ocean). An example of a primitive crust is on Vesta. ^{C22}	This crust rarely survives on larger bodies, and therefore is not expected to be observed on exoplanets.
Primary Crust	The crust that forms on a planet due to the differentiation and solidification of an initial magma ocean. While rare in the Solar System, there are examples of isolated regions on terrestrial bodies with primary crust: e.g., the lunar highlands, which formed from rocks (i.e., anorthosite) composed of plagioclase feldspars (a mineral group with the endmember minerals albite and anorthite) that crystallized in the magma ocean and floated to the top. ^{C22}	For young exoplanets, or ones without active geology, their crust might be primary. As an initially molten surface cools, dense condensed material that crystallizes out is sequestered deeper in the mantle (olivines, pyroxenes) while less-dense condensed material (like anorthosite) floats to the top. A similar analysis done by modeling the resulting top-level crust from cooling a compositionally specific magma ocean can be done to determine whether or not an observed crust is primary or not.
Secondary Crust	Produced from partial melts of solid mantle material (which are typically ultramafic rocks) that usually cools into mafic basalts. This form of crust is the most common on rocky planets in the Solar System, and replaces the primary crust over time due to mantle overturn and volcanism. On Earth, fresh secondary crust is formed at mid-ocean ridges. ^{C22}	Most exoplanets will probably have some form of secondary crust that has been formed from the cooled material produced via the partial melting of their solid mantle. Additional processes, like impacts, intense stellar irradiation, and past/present atmosphere might further alter the secondary crust. Exoplanets without active geology will have a more altered secondary crust.
Tertiary Crust	Requires the gradual modification of the secondary crust through a process like plate tectonics. The only known example is the continental crust of Earth. ^{C22}	Earth's thick and old tertiary crust is formed from plate tectonics, which are solely unique to Earth and require large amounts of surface water to occur (water lowers the melting point of mantle material, lowers viscosity, and reduces friction between plates (Tikoo & Elkins-Tanton 2017)). ^{Z16} While it is unknown whether or not observable exoplanets will have plate tectonics, it is possible that the idea of what constitutes a tertiary crust will expand with the as-of-yet unknown geologic processes that dominate in highly-irradiated, tidally locked rocky exoplanets.

Table A1 *continued*

Table A1 (*continued*)

Word (1)	Definition (2)	Exoplanet Context (3)
Surface Composition	The resulting surface composition of a terrestrial body, and therefore what can be detected directly with spectroscopy, is determined by a complex interplay of a planet’s initial formation composition, the cooling of the planet over time (which determines the level of differentiation and resurfacing processes that occur), the heating of the planet (which in the case of tidally-locked worlds with dayside temperatures above the melting point of the exoplanet’s crustal rock composition, can result in dayside magma oceans), whether or not the planet is still geologically active (i.e., Mars due its small size is now geologically dead), and subsequent alteration of the surface from a past or present atmosphere, liquid water, or impacts.	While these processes are difficult to disentangle from the detection of geology on a exoplanet, inferences can be made about what lead to a specific observed surface composition (see §5.1 and Table A3). For example, the detection of a granitic surface could indicate the presence of current or past Earth-like plate tectonics, the cooling of a planet that formed with a silica-rich mantle (an ‘exotic’ mantle), or the resurfaced metamorphic transformation of rock into a mineralogical form similar to granite (e.g., gneiss). The database presented in this work focuses on lab data of rocks with specific mineralogies and chemical compositions, where if the spectral signature of one is detected, how that rock (or something similar to it) is the primary spectral signature of an exoplanet’s surface can be deduced within the context of modern-day geology. Other works have explored the detection of an exoplanet’s chemical composition in lieu of a specific rock (SiO ₂ wt% content via detection of the Christiansen Feature (Paragas et al. 2025; First et al. 2025)), and methods by which to learn more about the surface composition from observed atmospheric compositions (Herbert & Sereinig 2025).
Resurfacing	The resurfacing of a planet via melted mantle material, and therefore the formation of secondary crust (which produces basalts in all the Solar System rocky planets) can occur in many different ways; on Mercury, the youngest surface material formed from lava flows generated by large bombardments, on Venus there is evidence of cyclic planet-wide resurfacing and active volcanism, on Earth there are plate tectonics and volcanism, on the Moon the dark basaltic regions were created by lava filling the lowlands, on Mars the northern lowlands are theorized to be fresh lava from a large impact while the southern highlands represent early explosive volcanism, and on Io there is sulfur-rich volcanism induced by tidal forcing from Jupiter. ^{Z16}	It is unknown which resurfacing processes are expected to dominate on large-size (super-Earths) and tidally-locked rocky exoplanets, though if the pattern in the Solar System is to be followed, it is inevitable that some amount of the surface area of any planet will be resurfaced with freshly cooled partial melts from the mantle (Meier et al. 2026).
Volcanism	The process by which interior material is brought to a planetary surface. Provides a mechanism by which mantle-like material ends up on the surface without processes like seafloor spreading or large impacts, and can also provide out-gassing of specific atmospheric species.	Volcanism has been predicted to be the source of atmospheric SO ₂ on some exoplanets (Bello-Arufe et al. 2025), where the amount of gas melted mantle material can hold is a strong function of its composition (Brugman et al. 2021). It is argued that in planets with high internal heat (i.e., due to tidal forcing) pipe volcanism akin to Io to dominate resurfacing (Reinhold & Schaefer 2024), while models of tidally locked planets predict nightside volcanism to dominate (Meier et al. 2026).

Table A1 *continued*

Table A1 (*continued*)

Word (1)	Definition (2)	Exoplanet Context (3)
Plate Tectonics	The process by which crustal material is reprocessed through subduction. In the Solar System this process only occurs on Earth and enables formation of rocks like granites. There are a myriad of other tectonic regimes (where Earth has a mobile lid that causes plate tectonics, and Venus has a proposed episodic lid that causes cyclical planet-wide resurfacing), which are described in Lyu et al. (2025) . The lack of plate tectonics is the stagnant lid regime, which can still have very active geology occurring (Noack et al. 2017).	As mentioned above in Tertiary Crust, plate tectonics are solely unique to Earth due to surface water, and therefore unlikely to be on hot observable exoplanet surfaces. Other forms of tectonism can still occur though, through processes like episodic tectonics on Venus. It is possible that partially molten lava exoplanets can have regions of solidified crust floating on top of the magma, which can undergo plate tectonic-like processes like subduction (Chao et al. 2021), or they have single-lid tectonics that recycle solid crust at the terminators (Meier et al. 2026).

NOTE—Broad geology definitions and their potential application to exoplanet studies. This table, which is in no way extensive, was generated to help guide future considerations for exogeology studies, foster collaboration between geologists and exoplanet scientists, and contextualize the categorization the surface albedos featured in the POSEIDON surface albedo database. ^{W14} indicates that information preceding the superscript was summarized from [Winter \(2014\)](#). ^{Z16} indicates that information preceding the superscript was summarized from [Zalasiewicz \(2016\)](#). ^{C22} indicates a crustal definition from [Condie \(2022\)](#).

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